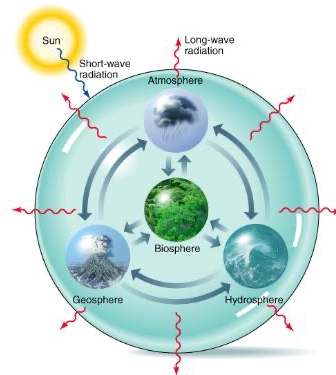


Introduction to Earth System

The components of the Earth System

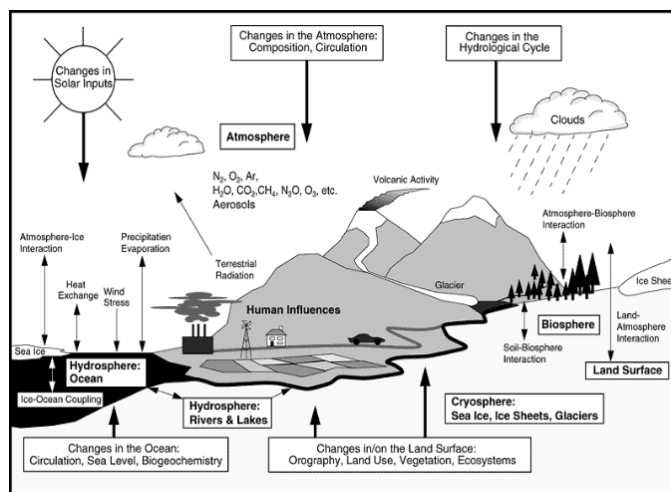


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Components of the Earth System

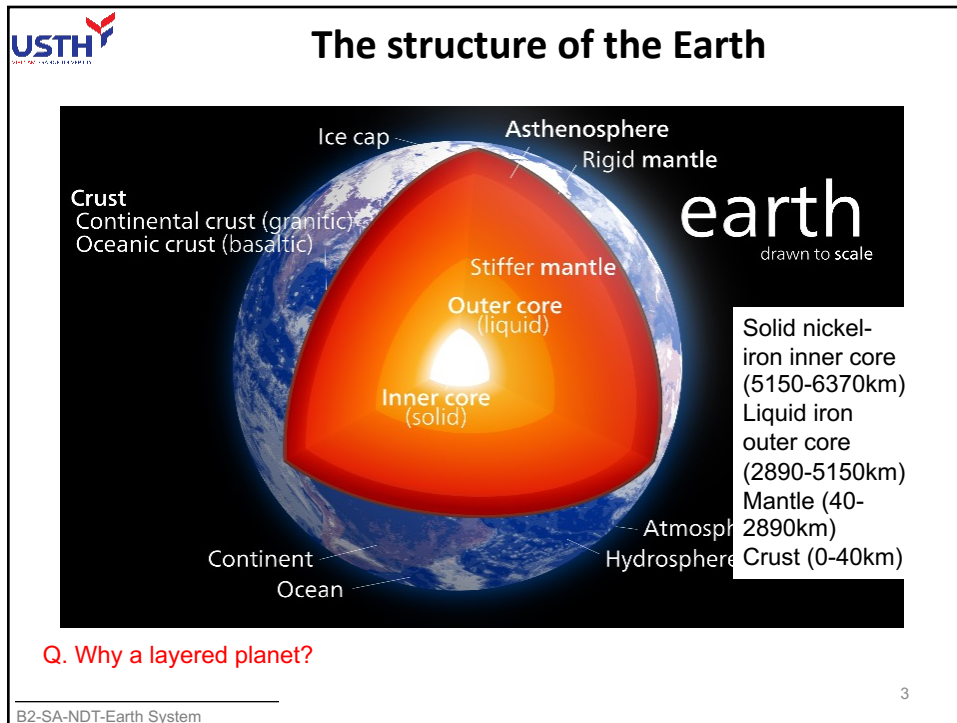
- Geosphere
- Atmosphere
- Hydrosphere
- Biosphere



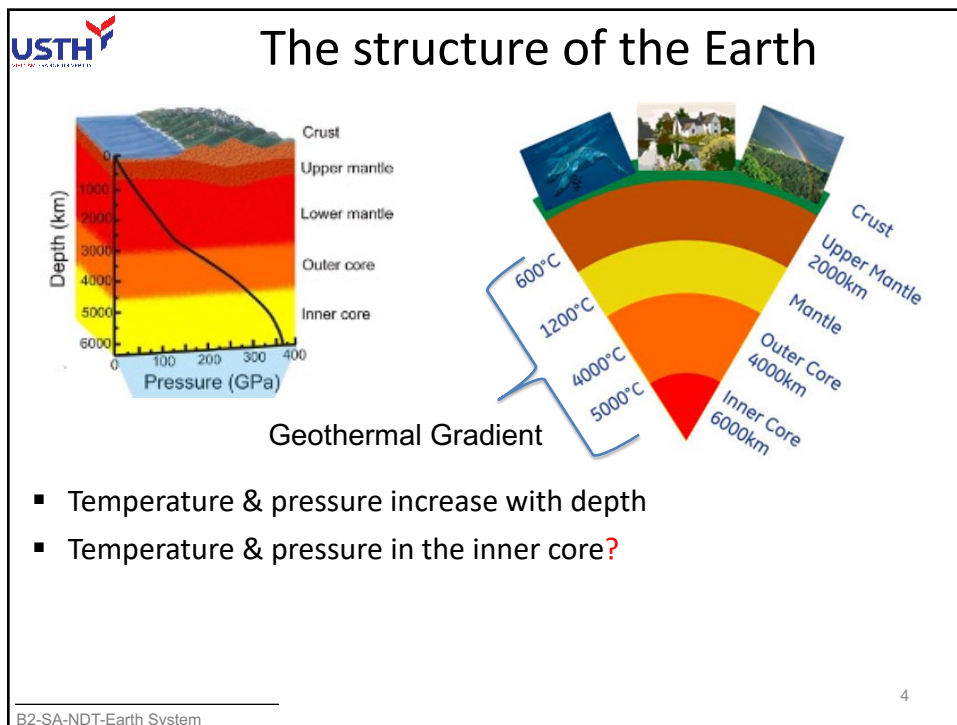
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IPCC, 2001

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4



Exercise #1

- Mass of the Earth: $M \sim 5.97 \times 10^{24} \text{ kg}$
- What is the average density of the earth?
- The density of surface rocks is $\sim 2700 \text{ kg/m}^3$
- what information you can obtain from the above result?

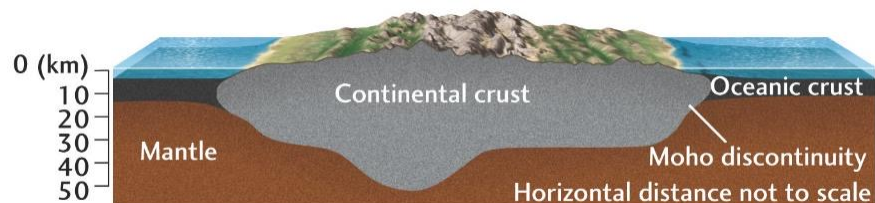
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5

5



The structure of the Earth

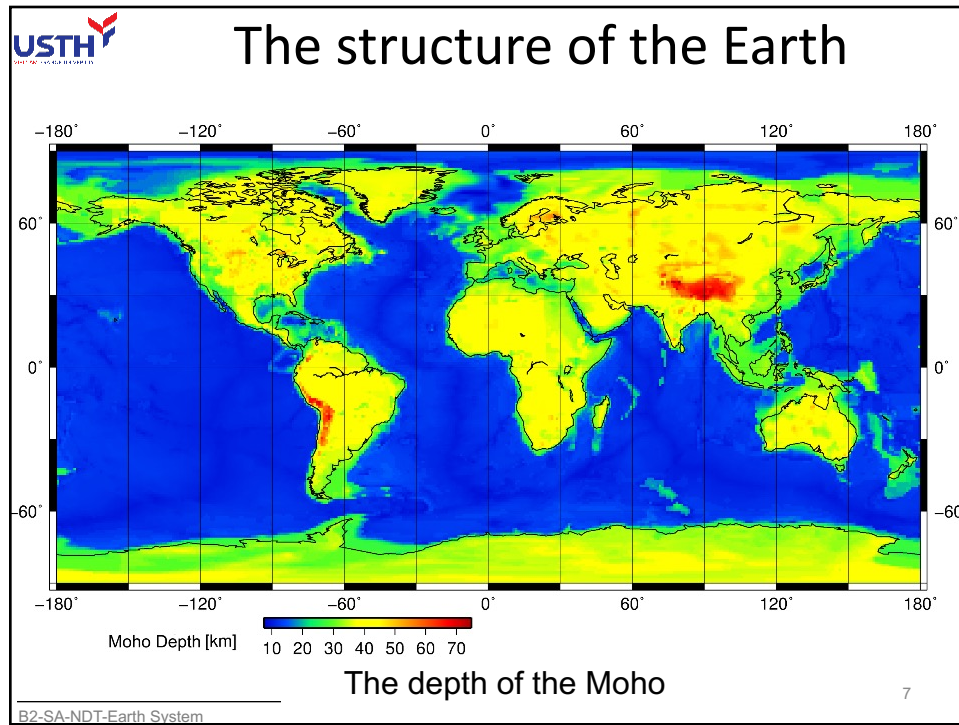


- The crust → the most familiar to us
- Rocks' age over oceanic crust and over continents?
 - <200 mil. yrs & can be > 3.5 bil. yrs, respectively
- ✓ Q: Oceanic crust or continental crust has a higher density?
 - **Oceanic crust** ← more iron and magnesium-rich minerals
 - The lower-density continental crust is thicker and stands higher
- Boundary between Earth's crust & mantle: rapid change in density → **Mohorovičić Discontinuity or Moho**

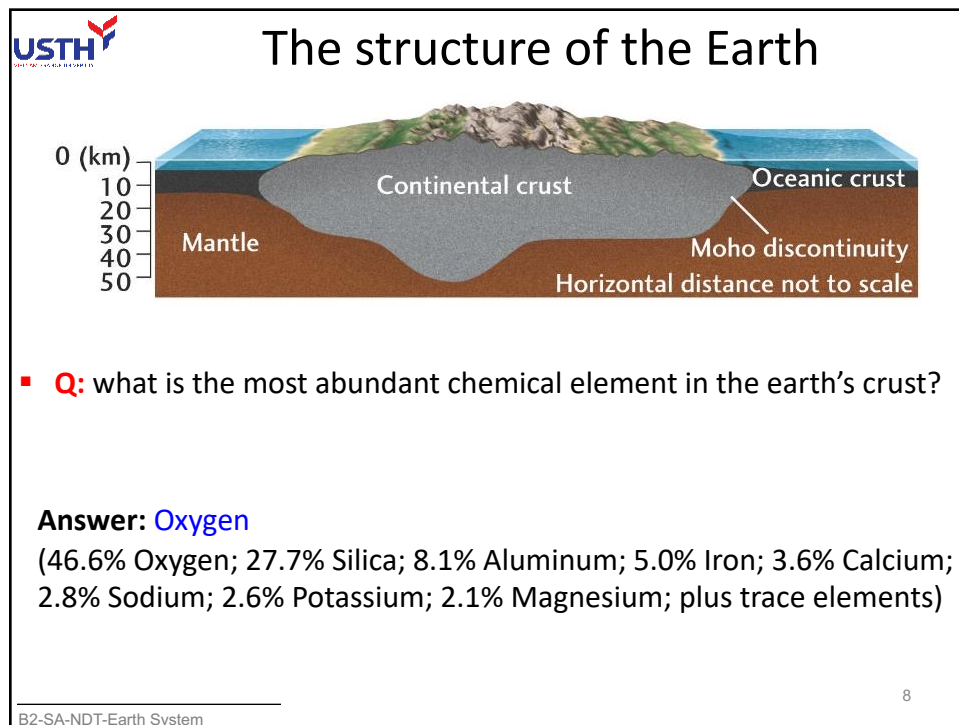
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6

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7



8

Plate Tectonics

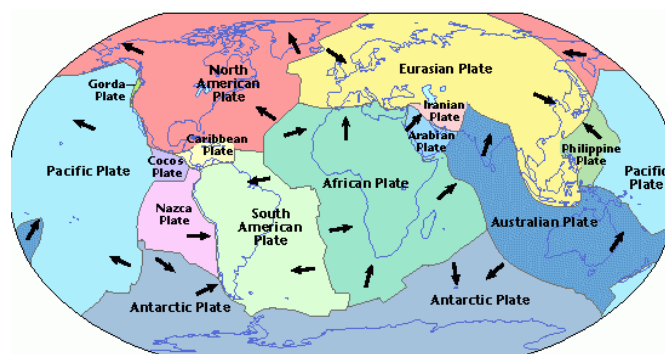
Q:

- Why are mountain ranges commonly near coastlines?
- What causes giant tsunami waves?
- Why are most devastating earthquakes near certain coastlines?

→ The dynamic of Geosphere

Plate Tectonics

- The Earth's crust → **major plates** moving in various directions → plate interaction
- "**Tectonic**" refers to the deformation of the crust as a consequence of plate interaction.

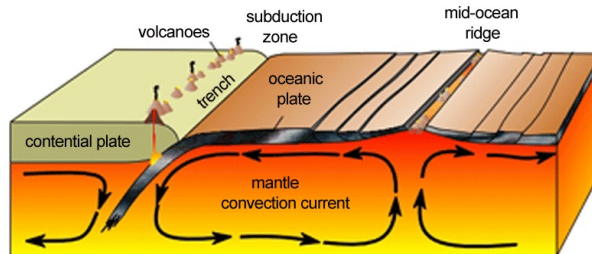


World plates (12 plates)



Plate tectonics

- Plates ← rigid lithosphere (made up of the crust and the extreme upper part of the mantle)



Mantle convection → transfer of thermal energy → causing the plates to MOVE

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11

11

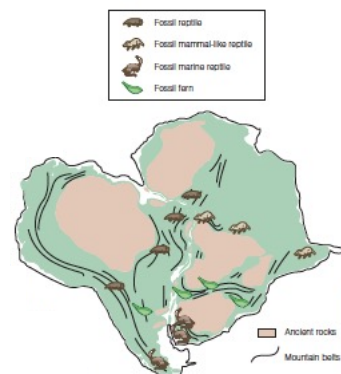


History



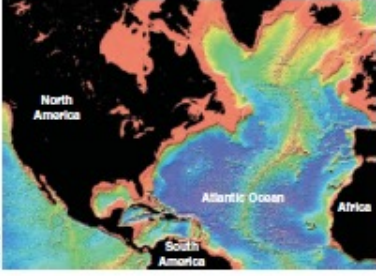
a few hundred million years ago → a supercontinent called **Pangaea**

- 1596:** Abraham Ortelius, a Dutch map maker suggested that Africa & South America were once connected
- 1912:** Alfred Wegener detailed the evidences → suggested that the continents drifted through the oceanic crust: **however, many scientists rejected this hypothesis**
- 1960:** the **seafloor spreading process** was confirmed



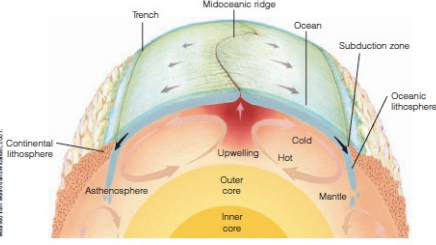
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12



Seafloor spreading

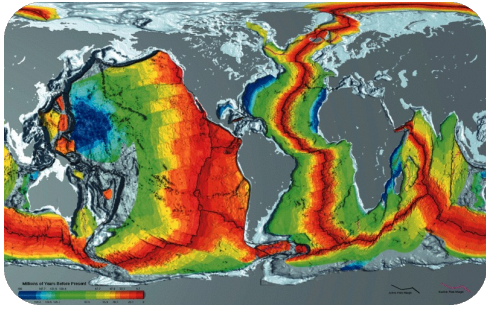
- Late 1940s → found the mid-oceanic ridge
- found that most earthquakes in the Atlantic Ocean were concentrated there
- 1960 → Harry Hess (Princeton Geologists) proposed the **seafloor spreading** process
- The estimated spreading rate in Atlantic (around equator) is ~ **2.5 cm/year**.



Q: How many years did it take for the Atlantic Ocean floor to form?

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13

13



Seafloor spreading

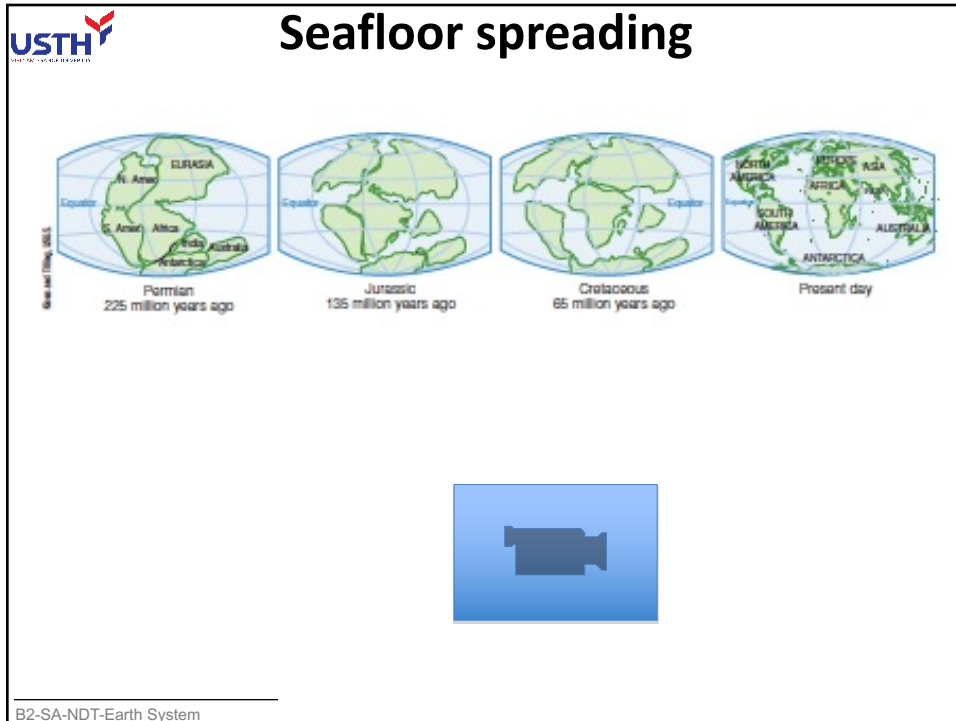
- Previous belief: ocean floor was extremely old → not correct
- At ridge axis:** youngest rock, thinnest crust, hottest flow

Evidences **support the modern theory of plate tectonics** (e.g. magnetic field, rock's age)

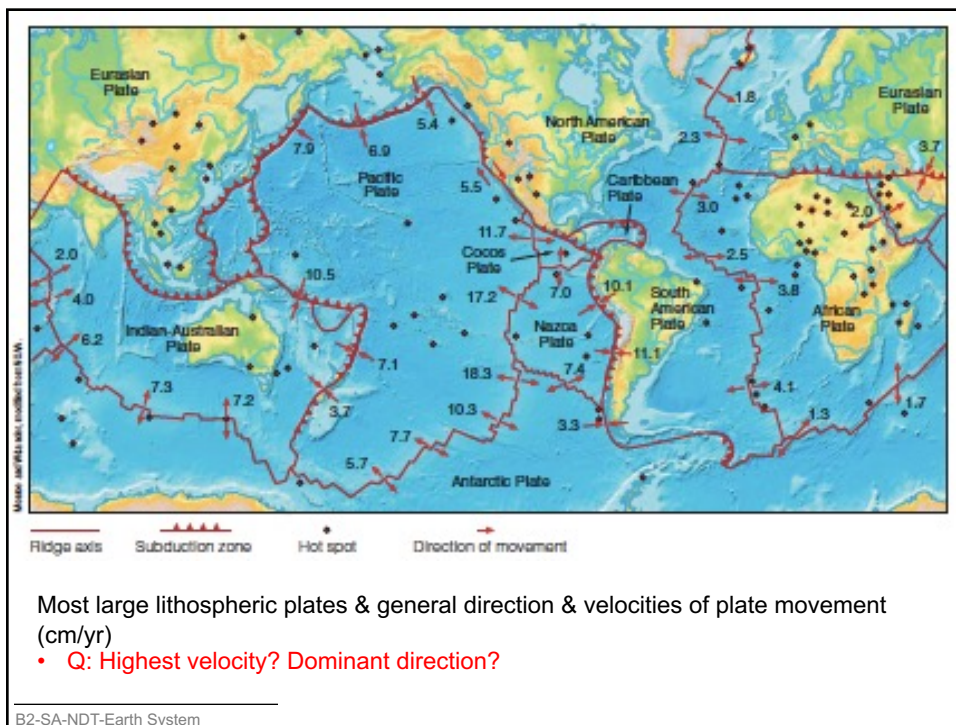
Q. A stripe of rock is 80 km away from the center of the Mid-Atlantic Ridge. Radiometric dating shows this rock is 2.5 million years old. What is the opening rate of the Atlantic Ocean?

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14

14



15



16

USTH

Plate boundaries

3 types of plate boundaries:

- convergent, divergent, & transform
- **Divergent boundaries** (e.g. ocean ridge)
- **Convergent boundaries**
 - ❖ One or both of the plates are oceanic lithosphere → the denser plate will slide down → **subduction zone**
 - ❖ Two continental plates → neither side is dense enough to be subducted deep into the mantle → **largest mountain ranges**
- **Transform boundary?** E.g. San Andreas Fault

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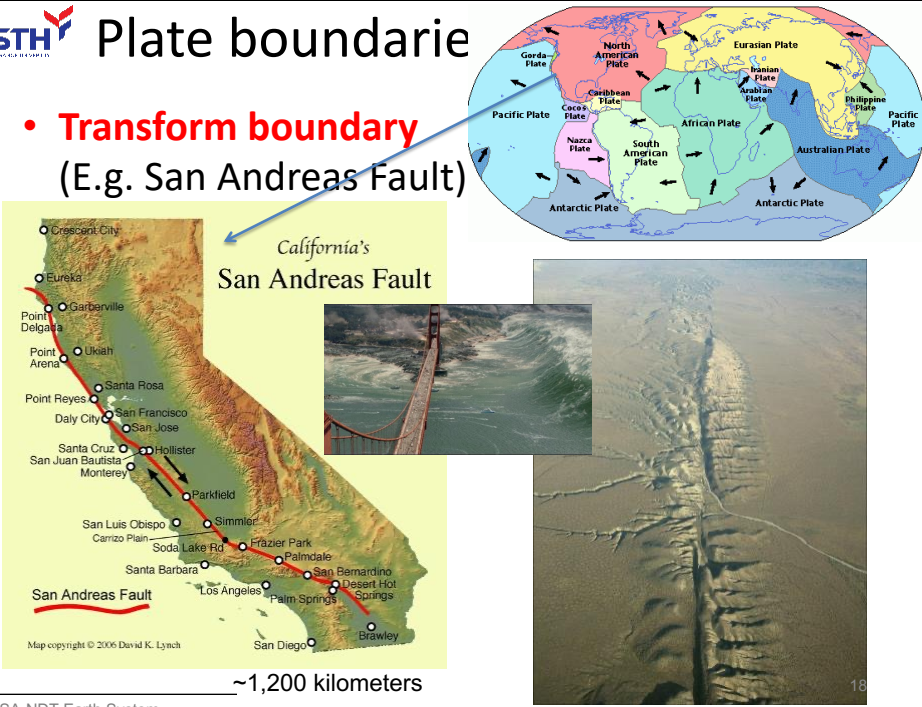
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USTH

Plate boundaries

- **Transform boundary** (E.g. San Andreas Fault)



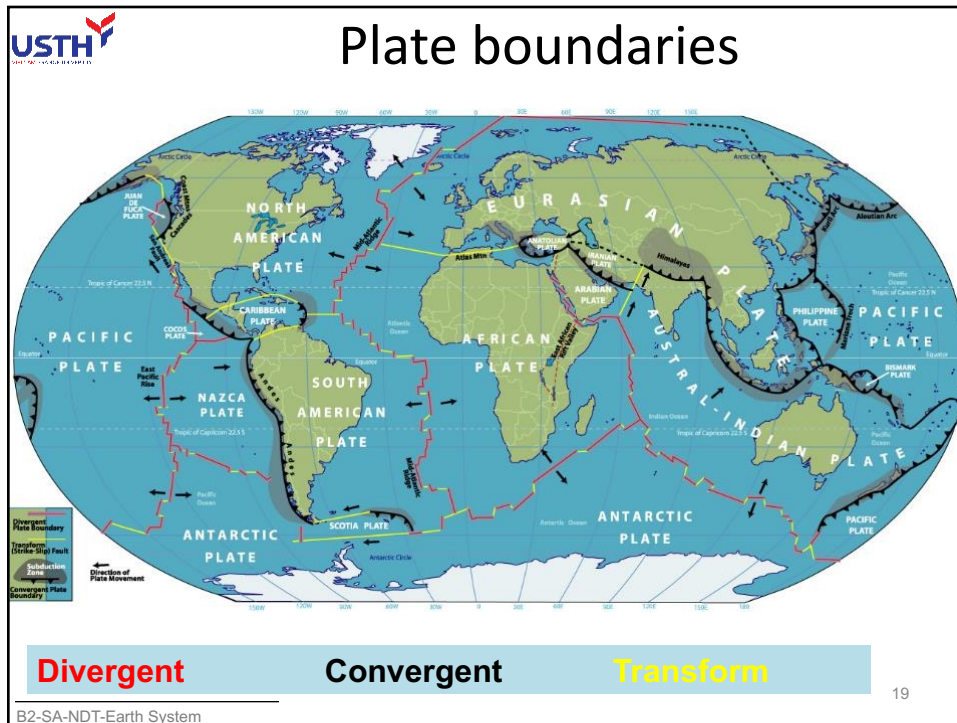
The composite image includes three main parts: 1) A world map showing the boundaries of major tectonic plates including the North American, Eurasian, African, Antarctic, Pacific, Indian, Australian, South American, Nazca, Caribbean, and Antarctic plates. 2) A detailed map of California highlighting the San Andreas Fault, which runs from the northern border near Crescent City down to the southern tip near Brawley. Major cities like San Francisco, Los Angeles, and San Diego are marked along the fault line. 3) An aerial photograph of the San Andreas Fault in a desert landscape, showing the characteristic stepped topography created by the fault's movement. An inset image shows the Golden Gate Bridge in San Francisco, illustrating the proximity of the fault to major urban centers.

California's San Andreas Fault

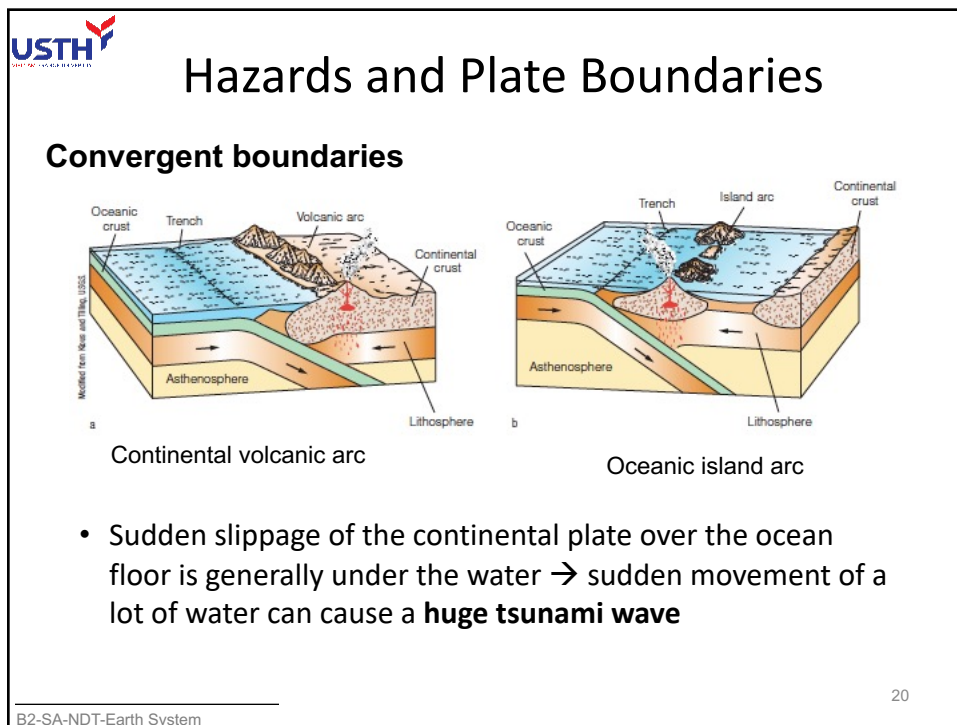
~1,200 kilometers

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18



19



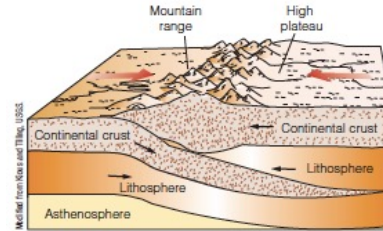
20



Hazards and Plate Boundaries

Collision of continents

- Neither plate sinks
- Promotes thickening of the combined lithospheres and growth of high mountain ranges
- Accompanied by large earthquakes



The Himalayas → created by collision between the Indian and Eurasian Plates

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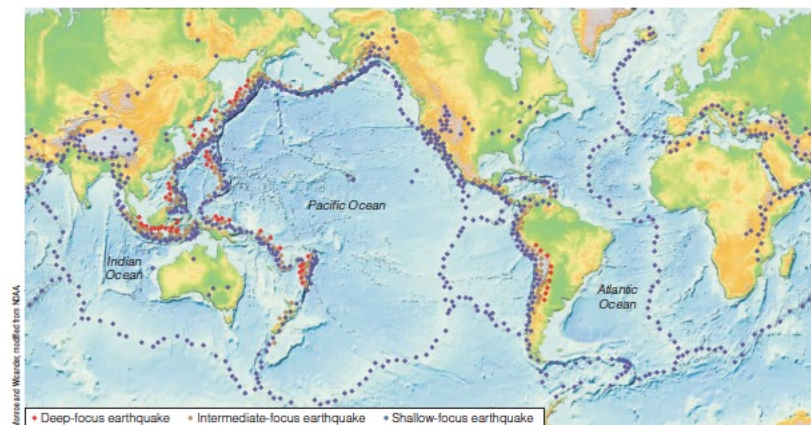
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21



Hazards and Plate Boundaries

- Most of Earth's earthquake & volcanic activity occurs along or near moving plate boundaries

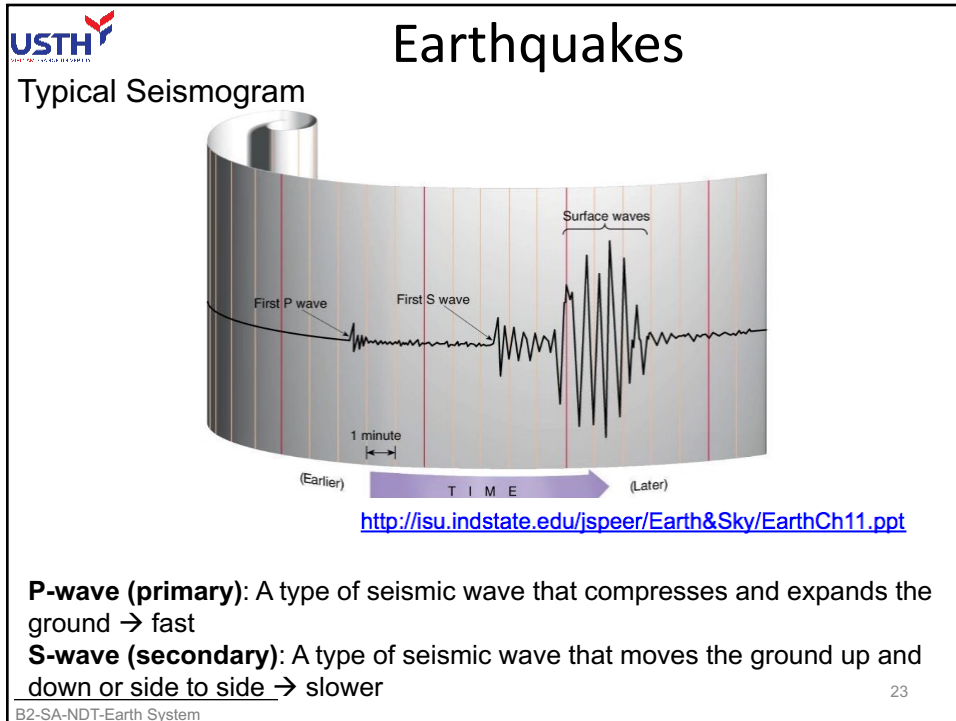


Earthquakes

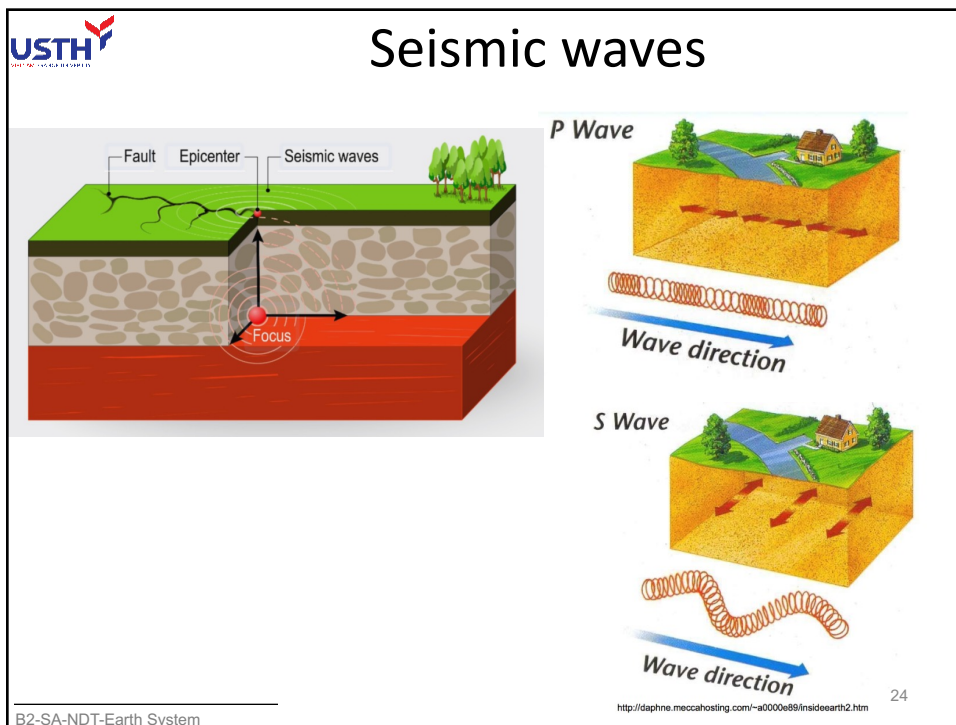
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22

22



23



24



Earthquakes → Seismology

Exercise:

Imagine a sudden shift along the San Andreas Fault. Three seismic stations A, B, and C record the arrival of the waves, with the following information:

Station	P-wave arrival time	S-wave arrival time	Δt (s)
Station A	10:00:00	10:00:12	12s
Station B	10:00:05	10:00:25	20s
Station C	10:00:02	10:00:17	15s

Assume that $V_p=8$ km/s, $V_s=4.5$ km/s

Locate the Epicenter of the earthquake?

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25

25



Hazards and Plate Boundaries

Volcanoes

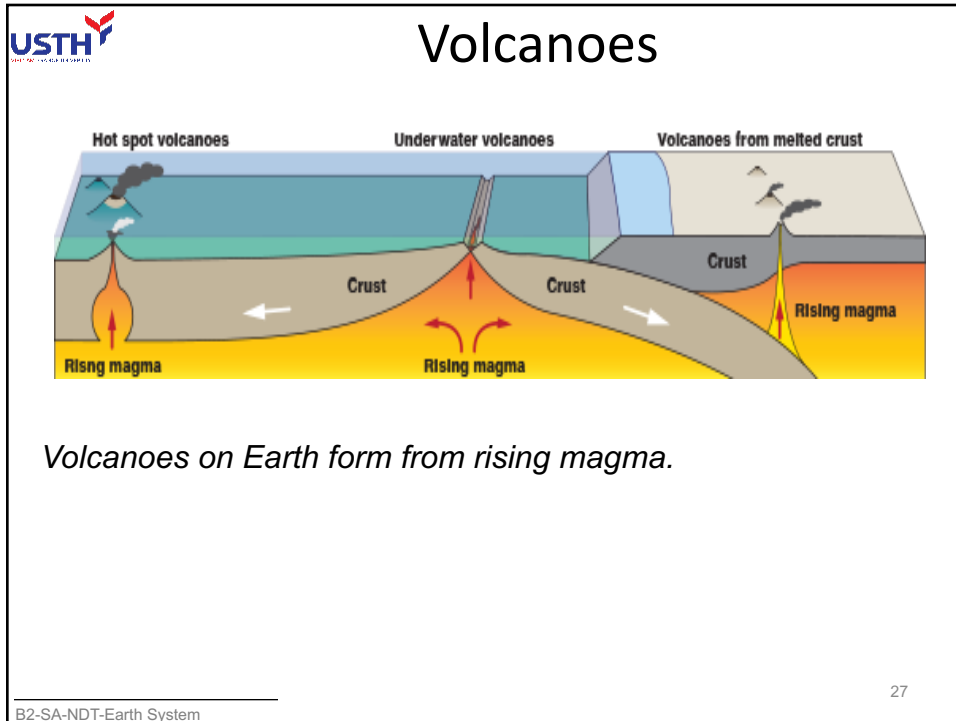


A **volcano**: an opening on the surface that allows material warmer than its surroundings to escape from its interior

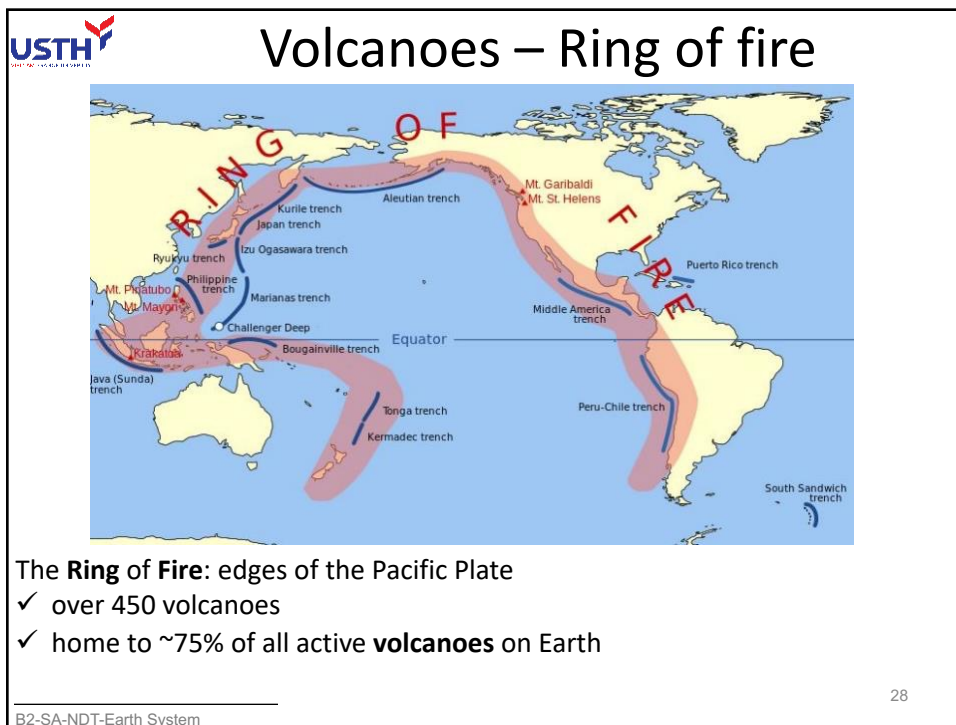
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26

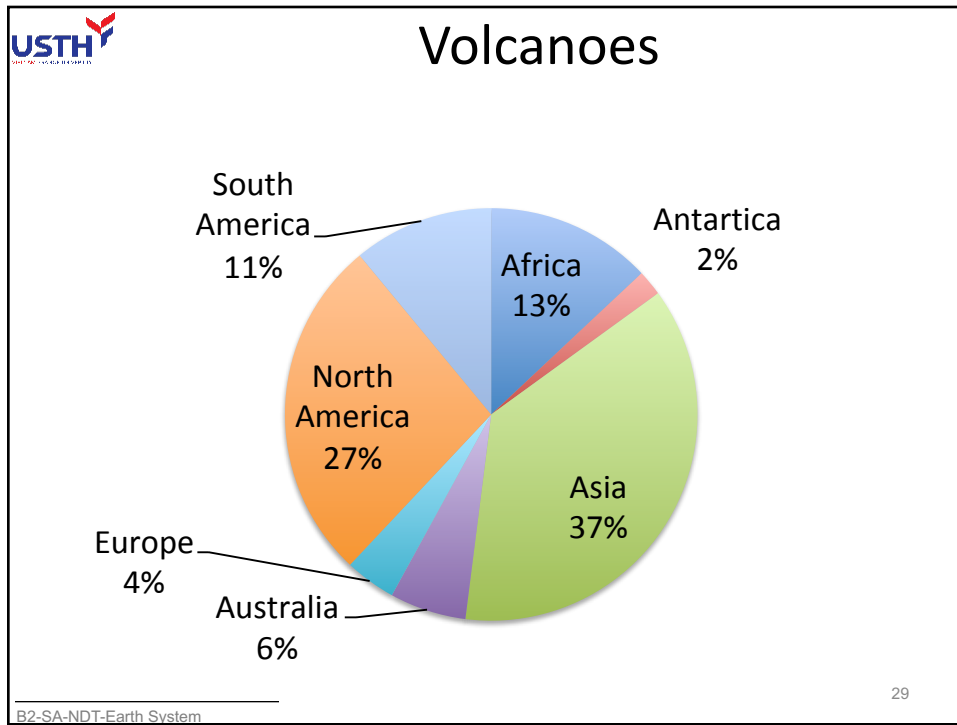
26



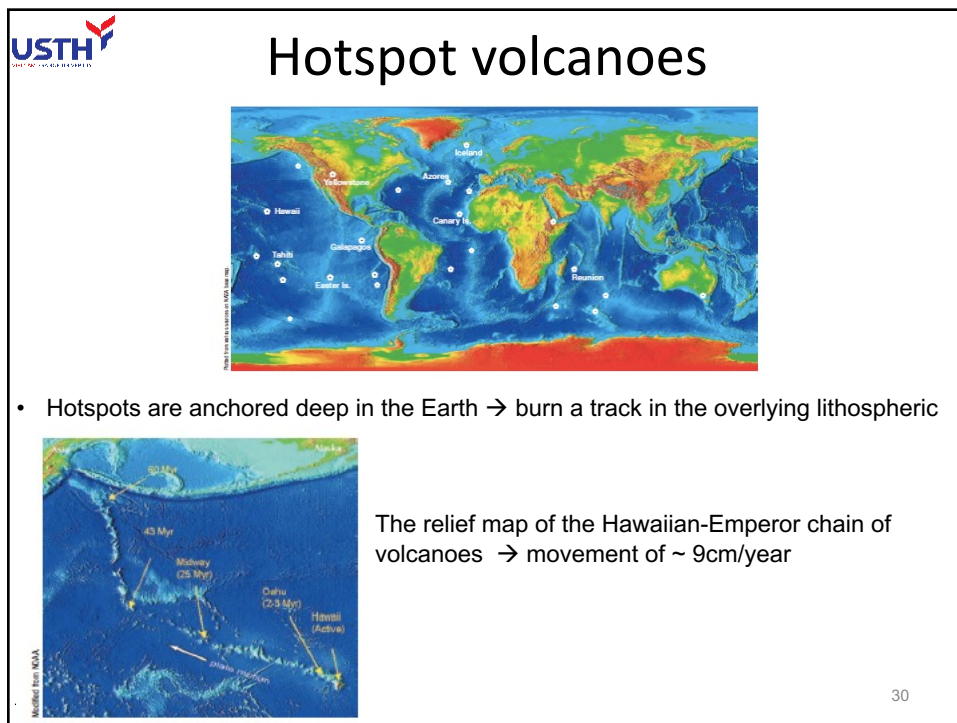
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28



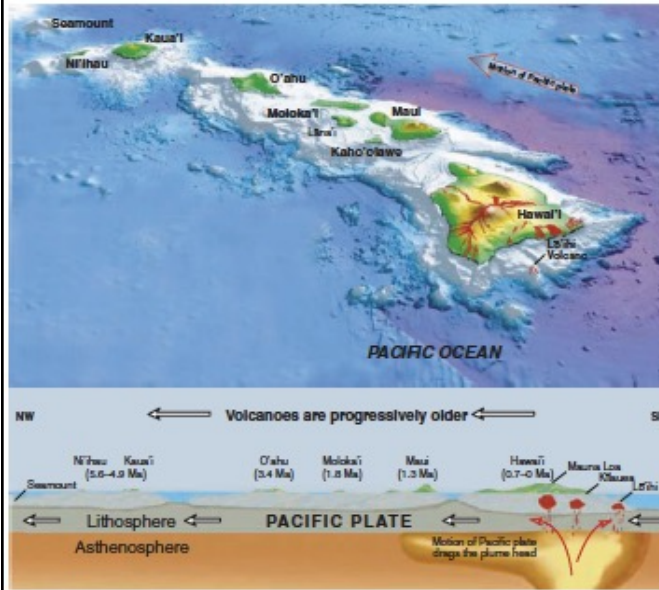
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30



Hotspot volcanoes



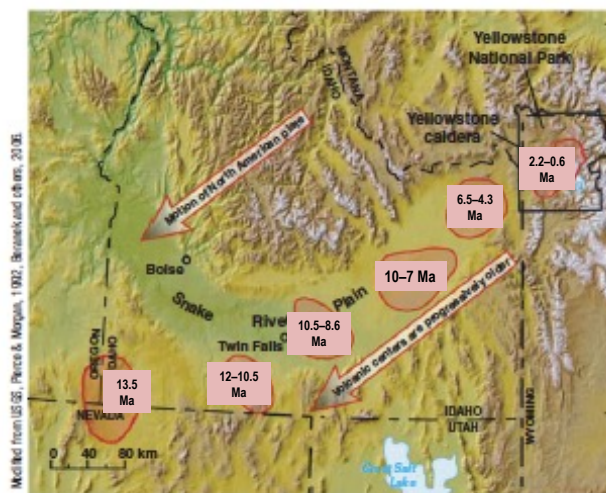
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31

31



Hotspot volcanoes



Snake River Plain of southern Idaho
(a series of extinct calderas)

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32

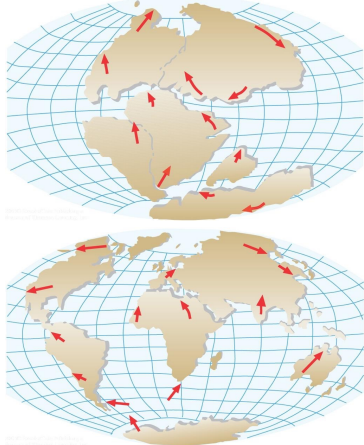
32

- The rising column or plume of hot rock appears to **remain fixed** in its place as one of Earth's plates moves over it
- a clear record of the **direction and rate of movement** of the lithospheric plates

Assuming that the hotspots remain fixed:

- Q: Estimate the rate of the movement of North American plate?
- Q: If the rate is changing?

USTH **How does tectonics influence Earth's climate?**



- Tectonic processes can have local to global-scale climatic impacts
- Direct & indirect impacts

Q. Compare the average temperature of an Earth with more polar continents covered by snow and ice vs. that of an Earth with a polar geography dominated by open water?

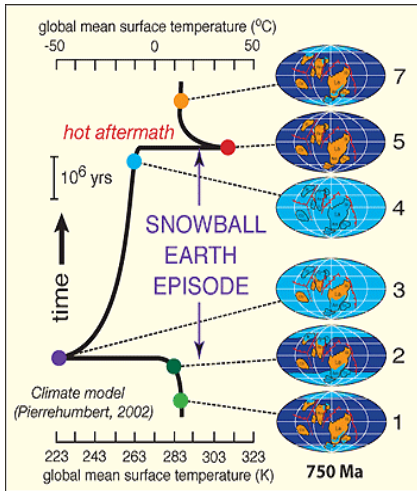
- Feedback process: ice-albedo
- Geographical boundary that prevent heat transfer from near-equator regions to higher latitudes
- High land elevation → favor colder climate

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33

33

USTH **Snowball Earth**



- 750, 635, 580 mil yrs ago: ice has covered the majority of the planet (average $T \sim -50^\circ\text{C}$)

Q. What caused the snowball earth?

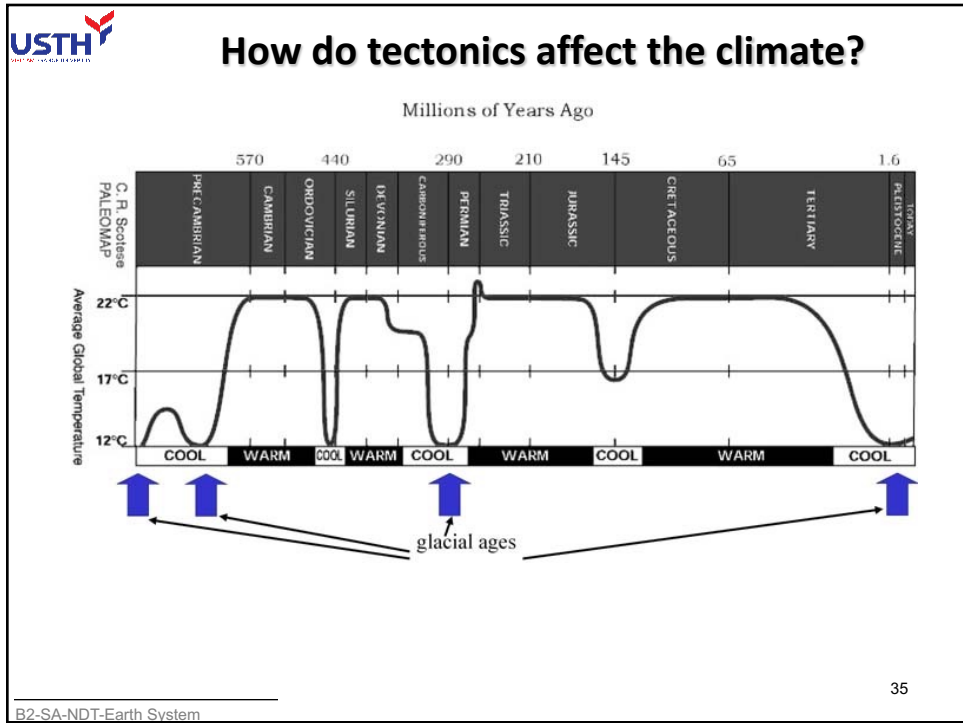
- Possibly by a lowering of atmospheric greenhouse gases through rock weathering, when the Sun was considerably dimmer than present

Q. However, given the ice-albedo feedback, how is it possible to escape a Snowball Earth?

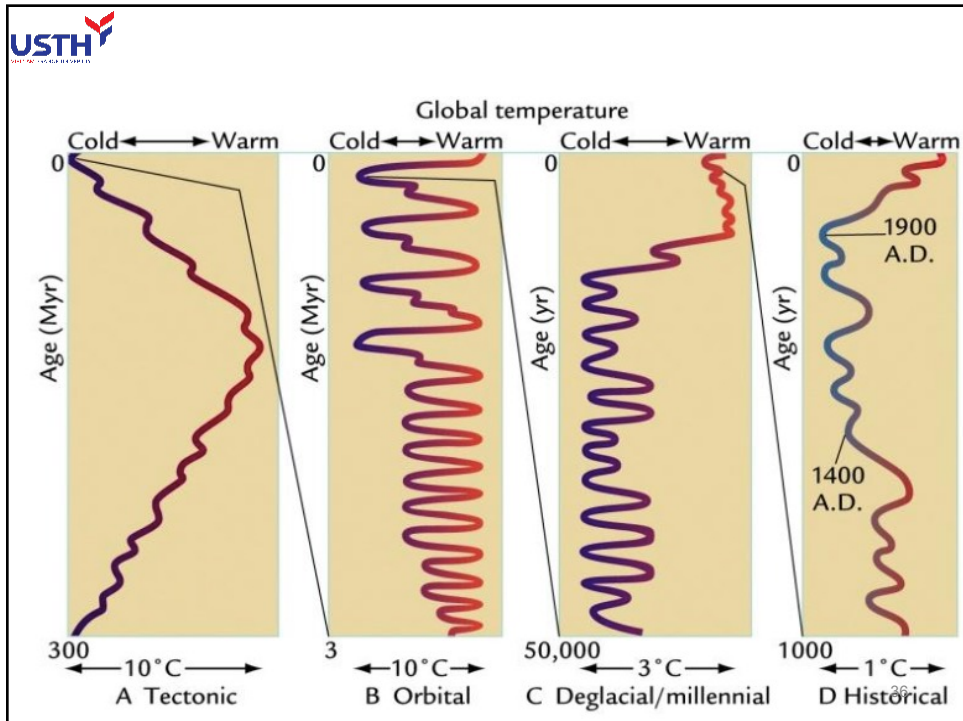
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34

34



35

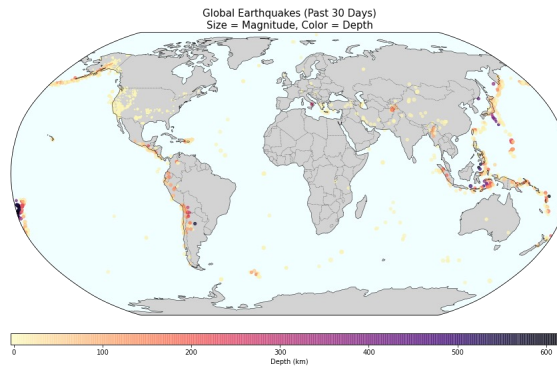


36



Practice #1

- Past 30-day earthquake data can be downloaded from:
https://earthquake.usgs.gov/earthquakes/feed/v1.0/summary/all_month.csv
(see <https://earthquake.usgs.gov/earthquakes/feed/v1.0/csv.php>)
- Plot the data on a world map using Python



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37

37